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Addressing fairness issues in deep learning-based medical image analysis: a systematic review

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Deep learning algorithms have demonstrated remarkable efficacy in various medical image analysis (MedIA) applications. However, recent research highlights a performance disparity in these algorithms when applied to specific subgroups, such as exhibiting poorer predictive performance in elderly females. Addressing this fairness issue has become a collaborative effort involving AI scientists and clinicians seeking to understand its origins and develop solutions for mitigation within MedIA. In this survey, we thoroughly examine the current advancements in addressing fairness issues in MedIA, focusing on methodological approaches. We introduce the basics of group fairness and subsequently categorize studies on fair MedIA into fairness evaluation and unfairness mitigation. Detailed methods employed in these studies are presented too. Our survey concludes with a discussion of existing challenges and opportunities in establishing a fair MedIA and healthcare system. By offering this comprehensive review, we aim to foster a shared understanding of fairness among AI researchers and clinicians, enhance the development of unfairness mitigation methods, and contribute to the creation of an equitable MedIA society.

The progressive advancement of artificial intelligence (AI) has garnered substantial attention and development in recent years, showcasing its efficacy in diverse practical applications such as autonomous driving, recommendation systems, and more. Notably, the data-centric approach inherent in AI methodologies has emerged as an indispensable asset in the domains of healthcare and medical image analysis.

Amidst the substantial research dedicated to refining the performance of machine learning (ML) or deep learning (DL) algorithms, a notable concern has surfaced among researchers. Although the performance of DL models may vary due to algorithmic factors such as random seed, some researchers witness a consistent performance fluctuation among patients with diverse characteristics or what is referred to as *sensitive attributes*. For instance, Stanley et al.¹ evaluated the disparities in performance and saliency maps in sex prediction using MR images, observing considerable discrepancies between White and Black children. Similarly, CheXclusion found that female, Black, and low socioeconomic status patients were more likely to be under-diagnosed, as compared to their male, White, and high socioeconomic status counterparts in chest X-ray datasets². This situation is not unique in medical image analysis (MedIA). Plenty of studies have

addressed the existence of unfairness across multiple imaging modalities (MRI^{3,4}, X-ray^{2,5,6}) and body parts (brain^{7–9}, chest¹⁰, heart^{3,4}, skin^{11,12}), and different sensitive attributes (sex^{9,13}, age^{14,15}, race^{16,17}, skin tone^{18,19}). Besides, this issue is also found in other healthcare applications where the inputs to the system are electronic medical records^{20,21} or RNA sequences²².

The phenomenon where the effectiveness of DL models notably favors or opposes one subgroup over another is termed unfairness²³. This issue is of profound ethical significance and necessitates urgent attention, as it contravenes fundamental bioethical principles²⁴. Moreover, it poses substantial impediments to developing reliable and trustworthy DL systems for clinical applications²⁵. One thing that must be taken in mind is that current studies about fairness mainly focus on the mathematical notions of fairness, i.e., using pre-defined metrics to measure the degree of unfairness. However, this form of definition does have shortcomings, as it can only deal with correlations rather than causation, leading to challenges in understanding how fairness statistics are derived and attributing responsibility²⁶. Besides, as the societal definitions of fairness evolve and are perceived differently by individuals, it is hard to find a proper formula to describe fairness consistently²⁷.

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Failure to adequately address fairness issues could result in a subgroup of patients receiving inaccurate or under-diagnoses^{2,28,29}, potentially leading to deterioration and causing lifelong harm to the patients. Clinicians may face difficulties in placing trust and confidently integrating deep learning methods into their routine practices. Recently, several studies in medical areas have urged the assessment of fairness in MedIA, including surgery³⁰, nuclear medicine³¹, and dental care³², which shed light on this area. Nonetheless, owing to distinct research focuses and various scopes in comprehending issues among AI scientists and clinicians, it is imperative to establish a bridge for understanding fairness between these two groups²⁵.

To this end, we have undertaken a systematic review aimed at addressing fairness concerns within DL-based MedIA. This review endeavors to introduce fundamental fairness concepts while categorizing existing studies about fair MedIA. Our aspiration is that this comprehensive review will aid both AI scientists and clinicians in understanding the present landscape and necessities concerning fair MedIA, thereby fostering the advancement of fair medical AI.

Results

The basics of group fairness

Fairness, as a concept describing collective societal problems, has been discussed widely throughout the development of human society²⁵. Although the definition of fairness varies in different areas, the gist is the same, i.e., all citizens should have the right to be treated equally and equitably. In AI research, fairness can be categorized into individual fairness³³, group fairness³⁴, max-min fairness³⁵, counterfactual fairness³⁶, etc. Among them, *group fairness* is used by most of the studies in DL-based MedIA. Thus, in this section, we present the basics of group fairness for better comprehension.

Group fairness requires that the DL model should have equal utilities for all the subgroups in the test set. Specifically, supposing a scenario where each subject in the dataset is a triplet, i.e., $S_i = \{X_i, Y_i, A_i\}$, where X_i denotes the image data of subject S_i , Y_i denotes the target label, and A_i denotes the *protected sensitive attributes*.

The general pipeline of group fairness evaluation is as follows. *First*, the test set is split into mutually exclusive subgroups by the sensitive attribute A_i . In MedIA, sensitive attributes can be information about the patients, including age, sex, race, skin tone, blood type, handedness, BMI, etc. *Then*, several group-wise fairness metrics are computed over each subgroup. The commonly used fairness criteria are shown in Table 1. For better understanding, we simulate four toy phenomenons where one of these fairness criteria is satisfied in Fig. 1. For example, as shown in Fig. 1a, ideal demographic parity requires the Male and Female groups to have equal probability of being predicted as illness, i.e., $P(\hat{Y} = 1|A = 0) = P(\hat{Y} = 1|A = 1)$. However, these fairness indicators could contradict each other and might not be satisfied at the same time. For example, as shown in Fig. 1a, the demographic parity is achieved. However, the accuracy of the Male group, i.e., $ACC_{Male} = P(\hat{Y} = Y|A = 0) = \frac{3+3}{3+2+2+3} = \frac{3}{5}$, while the accuracy of the Female group, i.e., $ACC_{Female} = P(\hat{Y} = Y|A = 1) = \frac{1+2}{1+3+4+2} = \frac{3}{10}$, which means that the Accuracy Parity is not satisfied. Thus, it is important to select proper fairness criteria based on specific tasks³⁷. *Finally*, the disparity between group-wise fairness metrics is computed to judge the overall fairness. The measurement function could be subtraction³⁸, division³⁸, SER³, STD³⁹, NR¹⁹, etc.

Compared to individual fairness³³, which requires similar outputs for similar samples and is easy to be affected by the outlier samples, group fairness stabilizes the fairness analysis by group-wise average measurements. But also has shortcomings as the model might satisfy fairness constraints in one grouping scheme while being unfair in another group scheme, resulting in an ambiguous conclusion. For example, Fig. 2 shows a situation where the patients can be split into subgroups based on *sex* and *race* (White/Black). Although DP is satisfied on *sex*, i.e., $P(\hat{Y} = 1|A = \text{Male}) = \frac{5}{10} = P(\hat{Y} = 1|A = \text{Female})$, DP is not satisfied with *race*, i.e., $P(\hat{Y} = 1|A = \text{White}) = \frac{6}{10} \neq \frac{4}{9} = P(\hat{Y} = 1|A = \text{Black})$.

A total of 687 papers were identified by our systematic research. After removing duplicates and irrelevant studies based on our criteria (see “Methods” section), 63 studies were included for information extraction and

Table 1 | Widely Used Criteria for Fairness

Metrics	Formula ^a	Explanation ^b
Demographic Parity (DP) ³³	$P(\hat{Y} = 1 A = 0) = P(\hat{Y} = 1 A = 1)$	The model outcome should not be affected by any sensitive attribute.
Accuracy Parity (AP) ¹¹⁸	$P(\hat{Y} = Y A = 0) = P(\hat{Y} = Y A = 1)$	The model should have an equal accuracy among subgroups.
Equalized Odds (EqOdds) ¹¹⁹	$P(\hat{Y} = 1 A = 0, Y = y) = P(\hat{Y} = 1 A = 1, Y = y), y \in \{0, 1\}$	The model should have an equal TPR and FPR among subgroups.
Equal Opportunity (EqOpp) ¹¹⁹	$P(\hat{Y} = 1 A = 0, Y = 1) = P(\hat{Y} = 1 A = 1, Y = 1)$	The model should have an equal TPR among subgroups.

^a $Y, \hat{Y} \in \{0, 1\}$ denotes the ground truth label and model prediction, respectively. $A \in \{0, 1\}$ denotes the sensitive attribute. Note that Y and A can be easily extended to multi-class situations.

^bTPR true positive rate, FPR false positive rate.

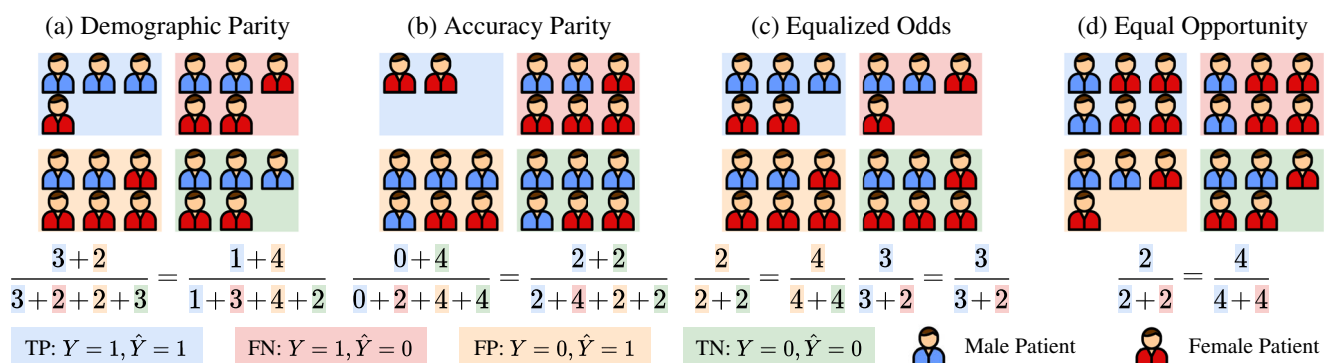


Fig. 1 | Ideal situations where various fairness criteria are satisfied. From left to right: **a** Demographic Parity, **b** Accuracy Parity, **c** Equalized Odds, **d** Equal Opportunity. The equations below compute the value of different criteria for the Male and Female groups.

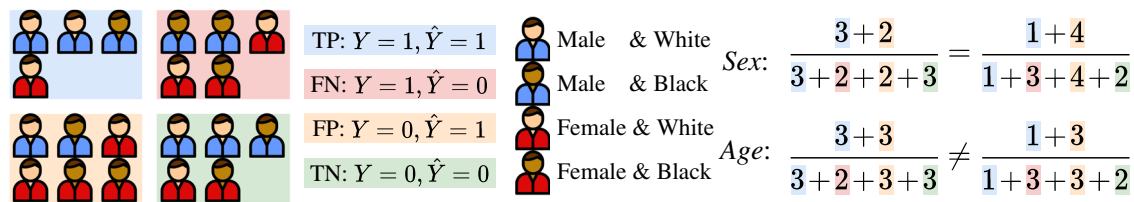


Fig. 2 | In a scenario involving two sensitive attributes, namely sex (male, female) and race (White, Black), demographic parity is achieved concerning sex but not race.

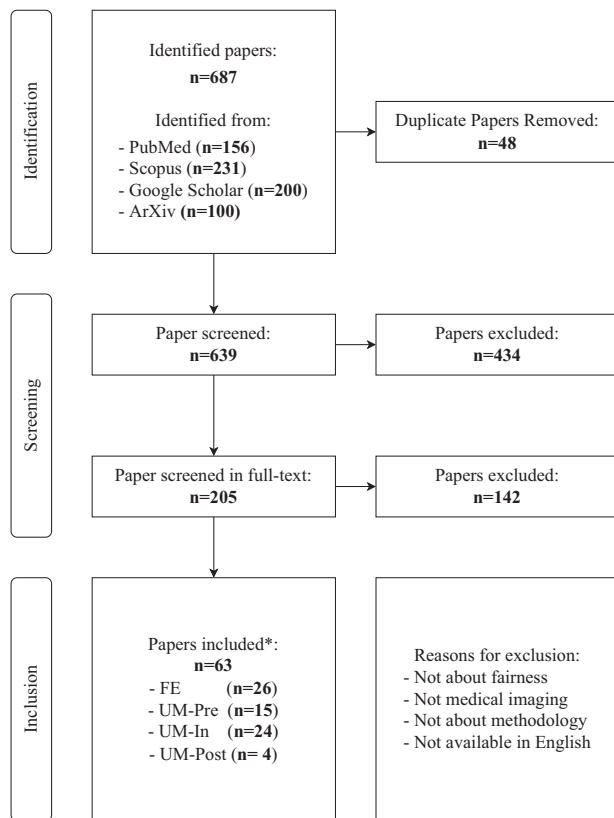


Fig. 3 | PRISMA diagram for this review. * denotes that six studies have been overcounted due to their involvement in research across multiple directions. FE fairness evaluation, UM unfairness mitigation, Pre pre-processing, In in-processing, Post post-processing.

categorization. Figure 3 presents the flowchart of this review based on PRISMA. Figure 4 and Table 2 describe the statistics of the extracted data.

From Figs. 3 and 4, we can find that research about fairness in MedIA began in 2019 and the annual number of publications grew about 6–7 per year. Fairness in MedIA is mainly assessed on Brain MRI, Dermatology, and Chest X-ray. This is easy to understand as the development of the brain is highly related to sex and age, while the skin part usually appears in dermatology images, which may lead to spurious relations for diagnosis. Chest X-ray, however, has the largest amount of samples compared to other modalities, which provides a potential for evaluating fairness. Besides, most of the current research focuses on classification and segmentation. Some attempts have also been conducted on anomaly detection⁴⁰ and regression¹². As for the sensitive attributes, sex, age, race, and skin tone are the most concerned.

Specifically, the research about fairness in MedIA mainly consists of two folds. One aims at benchmarking fairness in various medical tasks and discovering the mechanism behind unfair performances, and the other category attempts to mitigate unfairness in current applications. The studies

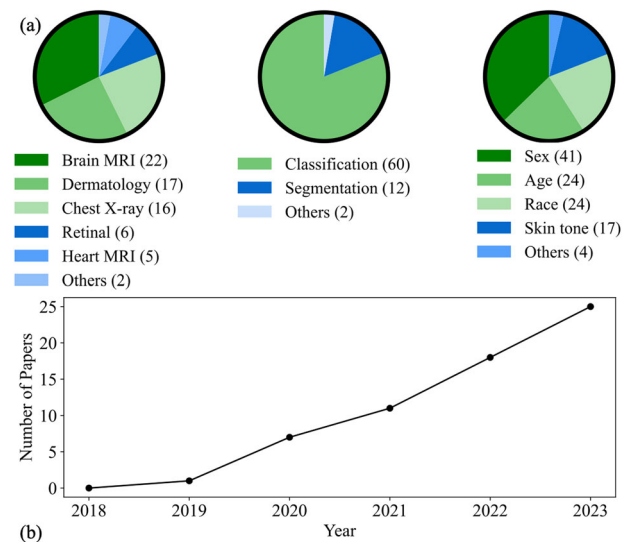


Fig. 4 | Summary of data extracted from studies in our systematic review. **a** Annual trends in research on fairness in MedIA. **b** Prevalence of various medical imaging modalities, research tasks, and associated sensitive attributes.

about fairness evaluation pay more attention to the existence of unfairness in various medical applications. While unfairness mitigation via in-processing strategies is more often considered (24/43). Note that some studies include research in multiple directions thus we repeat counts in Fig. 3.

Fairness evaluation

The starting point of addressing fairness issues in MedIA is to evaluate the existence of unfairness in MedIA tasks, by applying performance comparison among different sensitive groups. On the other hand, some researchers try to discover visual patterns that account for unfair model performance and study the relationship between fairness and other concepts.

Up to now, plenty of studies have been conducted on benchmarking unfairness in MedIA tasks. These studies evaluate diagnosis performance either on multiple network architectures or trained with different attribute ratios.

These studies vary in body parts and modalities, including brain MR^{1,7,41–43}, breast MR⁴⁴, Mammography⁴⁵, Chest X-ray^{2,5,6}, cardiac MR⁴, head and neck PET/CT⁴⁶, and skin lesion^{47,48}. Not surprisingly, most of the studies notice significant subgroup disparities in the utility except for the study in ref. 47, which does not find an observable trend between model utility and skin tone⁴⁸. Recently, researchers expanded the range of tasks and the type of algorithms, such as regression⁴⁹, anomaly detection⁴⁰, reconstruction⁵⁰, multi-instance learning⁵¹, random forest⁴⁴, and transformer⁵², which fills the gap in fair MedIA. Besides, Zong et al.⁵³ benchmark ten unfairness mitigation methods on nine medical datasets. However, they find that none of the ten methods outperforms the baseline with statistical significance. Similar results are also found in CXR-Fairness⁵⁴.

To go further, some research examines whether the subgroup ratio in the train set affects the performance of DL models. Larrazaba et al.¹³ train DL models on Chest X-ray datasets and find that the diagnosis performance

Table 2 | Overview of Studies in Fair Media

Research area ^a	Year	Citation	Imaging modality	Dataset	Task ^b	Sensitive attributes	Fairness metrics ^c
FE-benchmarking	2020	13	Chest X-ray	NIH Chest-XRay14, CheXpert	C	Sex	Δ AUC
	2020	5	Chest X-ray	MIMIC-CXR, ChestXray8, CheXpert	C	Age, Race, Sex, etc.	Δ AUC, Δ TPR
	2020	47	Dermatology	ISIC, SD-198	C	Skin tone	AP
	2021	54	Chest X-ray	CheXpert, MIMIC-CXR	C	Race, Sex	Δ AUC, Δ BCE, Δ ECE, Δ TPR, Δ TNR
	2021	2	Chest X-ray	MIMIC-CXR, CheXpert, ChestXray14	C	Age, Race, Sex	Δ FPR, Δ FNR
	2022	53	Multiple	Multiple datasets	C	Age, Race, Sex, Skin tone	Max-Min Fairness, Δ AUC, Δ BCE, Δ ECE, Δ TPR, Δ FPR, Δ FNR, EqOdds
	2022	8	Brain MRI	ADNI	S	Race, Sex	Δ DSC
	2022	9	Brain MRI	ADNI	C	Sex	Δ AUC, Δ ACC
	2022	4	Heart MRI	UK Biobank	S	Race, Sex	Δ DSC
	2022	1	Brain MRI	ABCD	C	Race	AP
	2022	7	Brain MRI	ABIDE	C	Sex, Sites	Δ AUC, Δ TPR
	2023	52	Brain MRI	UK BioBank	S	Race, Sex	Δ DSC, Performance Range, SER-DSC, STD-DSC, Bias Trend
	2023	48	Dermatology	ISIC	C	Skin tone	Δ BACC
	2023	6	Chest X-ray	CheXpert, MIMIC-CXR	C	Race, Sex	Δ TPR, Δ AUC
	2023	41	Brain MRI	ADNI	C	Sex, Race	DP, EqOpp, EqOdds
	2023	44	Breast MRI	TCIA	C	Race	AP
	2023	42	Brain MRI	ADNI	C	Age, Race, Sex	Δ ECE, Δ FPR, Δ FNR
	2023	46	PET/CT	HECKTOR	S	Age, Sex	Δ DSC, Δ TPR
	2023	43	Brain MRI	ADNI	C	Sex	Δ BACC
FE-discovery	2021	60	Mammography	DMIST	C	Race	Δ AUC
	2023	120	Chest X-ray	CheXpert	/	Race	
	2023	58	Multiple	Multiple datasets	C	Age, Race, Sex, Skin tone	Δ AUC, Δ ACC
	2023	59	Chest X-ray	NIH ChestX-ray8, CheXpert	C	Sex	Δ AUC
	2023	40	Brain MRI	ADNI	AD	Race, Sex	Δ MAE
	2023	57	Chest X-ray	CheXpert, NIH ChestX-ray 14	C	Sex	Δ AUC
	2023	12	Multiple	ISIC, BraTS, ADAS13	C, S, R	Age, Sex	Ap, Δ DSC, Δ RMSE
UM-pre-re-distribution	2021	3	Heart MRI	UK BioBank	S	Race, Sex	Δ DSC, SER-DSC, STD-DSC
	2023	61	Chest X-ray, Dermatology	JSRT, Stanford DDI	C, S	Age, Skin tone	Δ IOU, Δ ACC
UM-pre-harmonization	2019	62	Dermatology	ISIC	C	Skin tone	Δ AUC
	2020	63	Dermatology	ISIC	C	Skin tone	Δ AUC
	2021	65	Brain MRI	Multi-site MRI	C	Sites	AP
	2022	67	Dermatology	ISIC	C	Age, Sex	DP, EqOpp, EqOdds
	2023	66	Retinal	BRSET, Diabetes Center	C	Sex	Δ F1
	2023	64	Dermatology	Private	C, S	Skin tone	Δ AUC, Δ ACC
UM-pre-aggregation	2020	5	Chest X-ray	MIMIC-CXR, ChestXray8, CheXpert	C	Age, Race, Sex, etc.	Δ AUC, Δ TPR
	2021	69	Chest CT	CTPA	C	Age, Race, Sex	Δ ACC, Δ AUC, Δ TPR, Δ TNR, Δ PPV, Δ NPV
	2023	68	Brain MRI	iSTAGING, PHENOM, ABIDE	C	Age, Race, Sex	DP, EqOpp, EqOdds
UM-pre-synthesis	2020	70	Retinal	AREDS	C	Race	AP
	2021	71	Retinal	Kaggle EyePACS	C	Skin tone	Δ AUC, Δ AUC, Δ TPR, Δ TNR
	2022	19	Dermatology	Fitzpatrick-17k	C	Skin tone	EqOdds, NAR
	2023	72	Brain MRI	OASIS, UK BioBank	C	Age, Sex	Δ BACC, Δ TPR, Δ PPV

Table 2 (continued) | Overview of Studies in Fair Media

Research area ^a	Year	Citation	Imaging modality	Dataset	Task ^b	Sensitive attributes	Fairness metrics ^c
UM-in-adversarial	2020	73	Brain MRI, Bone CT	Private dataset	C	Age, Sex	Δ BACC, Δ TPR, Δ PPV
	2020	74	Dermatology	ISIC	C	Age, Sex	Δ AUC
	2021	15	Brain MRI	Private dataset	C	Age, Sex	Correlation
	2021	18	Dermatology	ISIC	C	Age, Sex, Skin tone	DP, EqOpp, EqOdds
	2021	3	Heart MRI	UK BioBank	S	Race, Sex	Δ DSC, SER-DSC, STD-DSC
	2022	75	Dermatology	ISIC	C	Skin tone	Δ AUC
	2022	78	Brain MRI	ABCD	C	Race	AP
	2023	77	Chest X-ray, Mammography	Private	C	Race	Δ TPR
UM-in-constraints	2021	10	Chest X-ray	CheXpert	C	Age, Sex	Δ AUC
	2022	79	Chest X-ray	MIMIC-CXR	C	Race, Sex	DP, EqOdds
	2022	80	Chest X-ray	MIMIC-CXR, ChestXray8	C	Sex, Sites	Δ AUC
	2022	121	Retinal	EyePACS	S	Skin tone	DP, EqOdds
	2023	81	Multiple	MIDRC, AREDS, OHTS, MIMIC-CXR	C	Age, Race, Sex	Relative Change
	2023	61	Chest X-ray, Dermatology	JSRT, Stanford DDI	C, S	Age, Skin tone	Δ IOU, Δ ACC
UM-in-disentanglement	2021	85	Brain MRI	HCP	C	Age	Δ AUC, Δ ACC, Δ F1
	2022	84	Brain MRI	USCF, ADNI, SRI	C	Age, Sex	Correlation
	2022	86	Brain MRI	UK BioBank, ADNI	C	Age	Δ MSE
	2022	83	Brain MRI	ABIDE	C	Age, Sex	AP
	2023	17	Chest X-ray	CheXpert	C	Age, Race, Sex	EqOdds, Δ AUC
UM-in-contrastive	2022	87	Dermatology	Fitzpatrick-17k, Stanford DDI	C	Skin tone	Division-ACC, DPM, EOM
	2022	19	Dermatology	Fitzpatrick-17k	C	Skin tone	EqOdds, NR-ACC
UM-In-MISC	2021	88	Dermatology	ISIC, Fitzpatrick-17k	C	Age, Sex	DP, EqOdds
	2023	11	Dermatology	Fitzpatrick-17k, ISIC	C	Sex, Skin tone	EqOdds, NAR
	2023	89	Multiple	Multiple datasets	C	Age, Sex, Skin tone	Δ AUC, Max-Min Fairness
UM-post-calibration	2023	61	Chest X-ray, Dermatology	JSRT, Stanford DDI	C, S	Age, Skin tone	Δ IOU, Δ ACC
UM-post-pruning	2022	79	Chest X-ray	MIMIC-CXR	C	Race, Sex	DP, EqOdds
	2022	90	Dermatology	ISIC, Fitzpatrick-17k	C	Sex, Skin tone	EqOpp, EqOdds
	2023	91	Brain MRI	ADNI	C	Age, Sex	Δ ACC, Δ AUC, Δ TPR, Δ TNR, Δ PPV, EqOdds

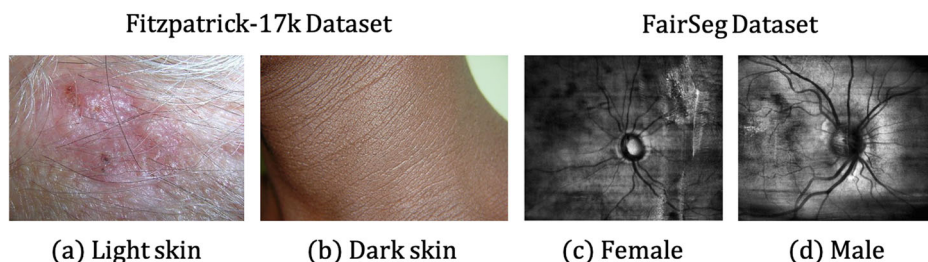
^aFE fairness evaluation, UM unfairness mitigation.^bC classification, S Segmentation, R Regression, AD anomaly detection.^cTPR true positive rate, TNR true negative rate, FPR false positive rate, FNR false negative rate, PPV positive predictive value, NPV negative predictive value, AUC area under curve, ACC accuracy, BACC balanced accuracy, BCE binary cross-entropy, ECE expected calibration error, MAE mean absolute error, IOU Intersection over Union, DSC Dice similarity coefficient, RMSE Root Mean Squared Error, DPM Division form of DP, EOM Division form of EqOpp.

between the female and the male is significantly different. This phenomenon could result from the fact that the diagnosis for one subgroup is more difficult than that for another subgroup, due to different image quality⁵⁵. Opposite to this finding, Petersen et al.⁵ evaluate the performance of DL models on the ADNI dataset with different subgroup ratios and notice that the performance of the under-represented group is not significantly different from that of the over-represented group. Besides, Ioannou et al.⁸ examine fairness by comparing the average Dice Similarity Coefficient (DSC) on each spatial location. Interestingly, they find that although unfairness exists in some regions severely, in other regions the DSC shows little or no disparity, which prompts us that unfairness evaluation should be conducted on different components respectively rather than on the overall utility.

Another category of studies focuses on unfairness source tracing and mechanism discovery. To discover the source of unfairness, many studies

focus on the visual disparities among samples with different attributes, which are recognized widely across different modalities⁵⁶, and suppose that unfairness comes from visual differences. For example, as shown in Fig. 5, there is a huge visual disparity between the White and the Black, the male and the female, and the young and the old. Liang et al. try to generate a sex-inverted X-ray image using a generative adversarial network and focus on the regions that have the largest disparity between the original image and its sex-inverted counterpart. By comparing the visual differences between the two images, they come up with some insights for understanding the source of unfairness and improving the DL models' interpretability. Similarly, Jimenez-Sanchez et al.⁵⁷ witness that the drain in the chest X-ray images may lead to shortcut learning for DL models, which causes unfairness. On the other hand, Jones et al.⁵⁸ notice that the ability of classifiers to separate individuals into subgroups is highly relevant to subgroup disparity. Besides, Kalb et al.⁴⁸ find that the level of unfairness varies when computing the

Fig. 5 | Visual disparities between images with different sensitive attributes. a, b images with dark skin and light skin from Fitzpatrick-17 Dataset¹⁶²; **c, d** images of a male patient and a female patient from FairSeg Dataset¹⁶⁹.



Individual Typology Angle (ITA) of the image using different methods, which proves that unfairness comes from the label annotation procedure. In chest X-ray diagnosis, Weng et al.⁵⁹ suppose that the differences in the imaging quality of the breasts between the male and the female are the reason for unfairness. Although the result does not support this hypothesis, their research gives insights for inspecting the source of unfairness. Moreover, Du et al.⁵⁰ evaluate model fairness in MRI reconstruction tasks and find that the estimated Total Intracranial Volume and normalized Whole Brain Volume might be the cause of unfairness.

Uncertainty also has a relationship with fairness. Lu et al.⁶⁰ try to compare several uncertainty measurements by evaluating subgroup disparity and find that although the aggregate utilities are similar, subgroup disparity varies among different uncertainty measurements. However, the result of ref. 12 conducted on three clinical tasks shows that although some unfairness mitigation methods have positive effects on fairness, they might harm the uncertainty of the model prediction.

Unfairness mitigation

According to ref. 11, strategies aiming at unfairness mitigation can be categorized into pre-processing, in-processing, and post-processing. The schematic diagram of each category is shown in Fig. 6.

The pre-processing methods mainly focus on data modification or remedy, which can be categorized into data re-distribution, harmonization, aggregation, and synthesis.

Re-distribution addresses unfairness by adjusting the balance of subgroups. This can be achieved by either resampling the train set or by controlling the number of samples in different subgroups within each mini-batch. By ensuring a more balanced representation of subgroups, re-distribution can help to mitigate unfairness in DL models. Puyol-Antón et al.³ adopt stratified batch resampling on the baseline model and notice a significant improvement in fairness. This strategy is also proved to be efficient in ref. 61.

Data harmonization removes sensitive information from the input images. This can be conducted by using segments or bounding boxes to remove the skin part from dermatological images^{62,63}, enhancing lesion boundary using image processing method⁶⁴, applying Z-score normalization or ComBat algorithm⁶⁵, or using differential privacy methods⁶⁶. Recently, Yao et al.⁶⁷ use a language-guided sketching model to transform the input images into sketches. Their result shows less disparity among subgroups, which indicates the potential of data harmonization in mitigating unfairness.

Data aggregation mitigates unfairness using information introduced by external datasets. The external dataset could be in the same modality^{5,68} or different modalities⁶⁹. For example, Zhou et al.⁶⁹ adopt the information of electronic health record data via an ElasticNet and improve the fairness of the model on pulmonary embolism detection by multi-modal fusion.

Data synthesis uses generative models to synthesize new samples to increase the number of training samples and balance the subgroup ratio in the training set. This can be done by either generating new samples with the same sensitive attribute as the original samples while varying in target label⁷⁰ or by generating new samples with opposite sensitive attributes but preserving the target label^{19,71,72}.

The in-processing methods focus on adjusting the architecture of models or adding extra losses or constraints to reduce biases among subgroups.

The adversarial architecture is the most common in-processing method, which adds an adversarial branch to the original architecture to minimize the influence of sensitive information in the latent space by using a gradient reversal layer as the adversarial branch^{3,15,21,73–78}. The biggest difference between them is the choice of loss functions for sensitive attribute prediction. One step further, Li et al.¹⁸ add an extra branch except for the adversarial one, which predicts the degree of fairness on the test set without knowing the sensitive attributes.

Another category of in-processing methods directly adds fairness-related constraints to the optimization objective. The constraints include GroupDRO⁶¹, differentiable proxy functions of fairness metrics^{10,79}, bias-balanced Softmax⁸⁰, and margin ranking loss⁸¹. However, as shown in¹⁰, this type of method can lead to over-parametrization and overfitting, resulting in a fluid decision boundary that can lead to fairness gerrymandering⁸², i.e., the DL models might be fair on *sex* and *age*, respectively, but are unfair when evaluated on the combination of *sex* and *age*.

Disentanglement learning extracts the feature vector from the input image and projects the feature vector into a task-agnostic portion and a task-related portion. This can be achieved by either maximizing the entropy of the task-agnostic portion⁸³ or by maximizing the orthogonality between the task-agnostic and the task-related portions¹⁷. Some studies regard the relationship between the two portions as a linear model. For example, Vento et al.⁸⁴ introduce the metadata into the model and describe the relationship using a general linear function, i.e., $\mathbf{f} = \mathbf{M}\beta + \mathbf{r}$, where $\mathbf{f}$ is the origin feature vector, $\mathbf{M}$ is the metadata, β is the learnable coefficient, and $\mathbf{r}$ is the residual task-related portion. A similar method is also used in⁸⁵ to mitigate the confounding bias in an fMRI dataset, where the coefficient β is estimated either on the whole dataset or each fold of the training set. The relationship could also be simply joint, as shown by Aguila et al.⁸⁶, who train a conditional variational auto-encoder on structural MRI data to disentangle the effect of covariates from the latent feature vectors.

The intuition behind contrastive learning is that in the latent feature space, the distances among feature vectors belonging to the same target class and different sensitive attributes should be minimized, whereas the distances among feature vectors from different target classes and the same sensitive attribute should be maximized. For example, Pakzad et al.¹⁹ use a skin tone transformer based on StarGAN to transform the skin tone of the input image and then regularize the L_2 norm between the feature vectors extracted from the shared feature extractor. Furthermore, Du et al.⁸⁷ implement a contrastive learning schema by adding an extra contrastive branch constraining the projected low-dimension feature vector.

There are some other attempts to mitigate unfairness in the in-processing procedure. Inheriting the idea of domain adaptation, FairAdaBN¹¹ reduces unfairness by adding extra adapters to the original model which can adaptively adjust the mean and variance of the feature vector according to the sensitive attribute. Fan et al.⁸⁸ design a special federated learning setting, where each client in the swarm learning only consists of samples with the same sensitive attribute. Moreover, FairTune⁸⁹ tries to mitigate unfairness by using parameter-efficient fine-tuning from large-

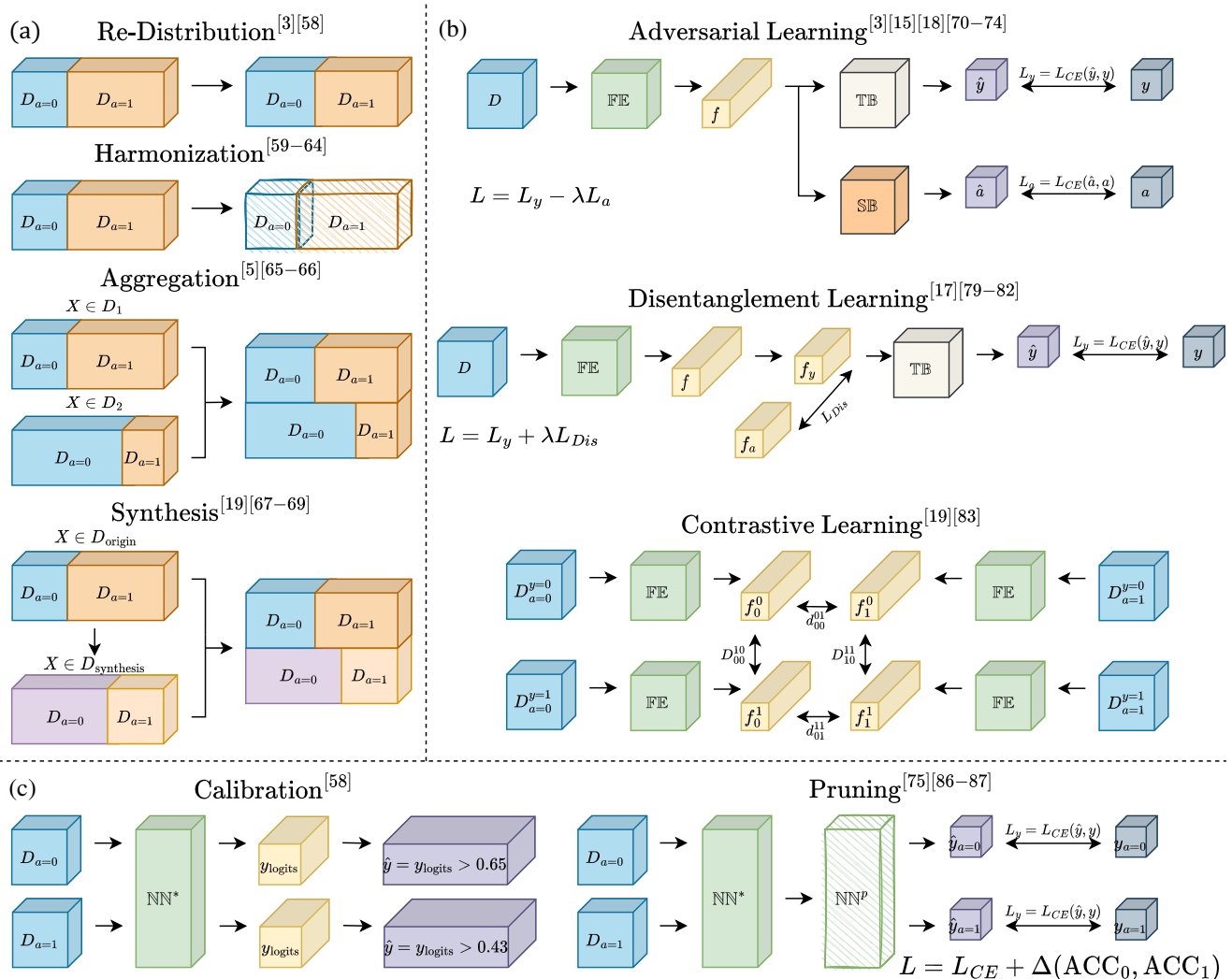


Fig. 6 | Schematic diagram of unfairness mitigation algorithms. a Pre-processing methods. D_1, D_2 : Two independent datasets; $D_{\text{origin}}, D_{\text{synthesis}}$: The original dataset and synthesized dataset. **b** In-processing methods. FE, TB, SB: Feature Extractor, Target Branch, and Sensitive Branch, which are three parts of an adversarial network; f : latent feature vector; $y, a, \hat{y}, \hat{a}$: the ground truth target task label, sensitive attributes, and their corresponding predictions generated by the neural network; L_{CE} : Cross-Entropy loss, measuring the difference between the predicted label and

the ground truth label; L_{Dis} : Disentanglement loss, for example, MMD-Loss¹⁷⁰, measuring the distance between two distribution; D_{00}^{10} : requiring the maximum distance between f_0^0 and f_1^1 ; d_{00}^{01} : requiring the minimum distance between f_0^0 and f_1^1 . **c** Post-processing methods. NN^* : a pre-trained and fixed Neural Network; y_{logits} : predicted probability of y , range from 0 to 1; NN^P : pruned NN^* ; $\Delta(\text{ACC}_0, \text{ACC}_1)$: difference between accuracy on subgroup test set $D_{a=0}$ and $D_{a=1}$.

scale pre-trained models. By considering fairness, it can improve fairness in a range of medical image datasets.

The post-processing methods mitigate unfairness by processing the fixed models by calibrating the output of DL models or pruning the model's parameters.

Calibration uses different prediction thresholds for each subgroup to achieve fairness. Oguguo et al.⁶¹ apply the reject option classification algorithm, which believes that unfairness occurs around the decision boundaries. It sets an outcome-changing interval where the output of the unprivileged group is re-modified. The calibration method is one of the most useful mitigation methods as it can be easily adopted on multiple DL models.

Pruning tries to satisfy fairness criteria by distilling the model's parameters. In their recent work, Wu et al.⁹⁰ compute the saliency of each neuron in the network and utilize a pruning strategy to remove features associated with a specific group. This approach helps prevent sensitive information from being encoded into the network. Moreover, Marcinkevics et al.⁷⁹ also use a pruning strategy to deal with unfairness, where the sensitive attribute is unknown. To go further, Huang et al.⁹¹ extend the neuron importance

measurement proposed in ref. 79, achieving better fairness while reserving model performance.

Fairness datasets in MedIA

To prompt fairness research in MedIA, we collect publicly available datasets with sensitive attributes in Table 3, which are categorized by image modality type, task type, attributes type, and the number of images in each dataset. We hope this Table can benefit the late-comer to find proper data to evaluate their algorithms Table 4.

Discussion

When adopting AI algorithms in medical applications, clinicians must be aware that, during the development of AI algorithms, most of the methods only focus on the diagnosis performance while ignoring whether the algorithm has biased or unfair utilities towards different subgroups. As a result, clinicians may find that the algorithm tends to underdiagnose specific sub-populations, and the corresponding treatment for each group might also differ. This phenomenon will make the clinician confused about clinical decision-making and wonder why the AI algorithms perform unfairly like that.

Table 3 | Available Medical Datasets for Fairness Assessment

Image modality	Dataset	Task ^a	Sensitive attributes								# Images
			Age	Sex	Race/ Skin tone	Marital	Drink	Smoke	Body prams ^b	Handness	
Chest X-ray	CheXpert ¹²²	C, D	✓	✓	✓						224,316
	NIH Chest X-ray ¹²³	C, D	✓	✓							112,120
	MIMIC-CXR ¹²⁴	C, D	✓	✓	✓	✓					371,858
	PadChest ¹²⁵	C	✓	✓							160,868
	BrixIA ^{126,127}	C	✓	✓							4703
	JSRT ¹²⁸	C	✓	✓							154
	COVID-ChestXray ¹²⁹	C, S	✓	✓							950
	Montgomery County X-ray ¹³⁰	C, S	✓	✓							138
	Shenzhen Hospital X-ray ¹³⁰	C	✓	✓							662
	NIH PLCO ¹³¹	C	✓	✓	✓						205,000
	VinDr-CXR ¹³²	C	✓	✓							18,000
Chest CT	NSCLC ¹³³	S	✓	✓							422
	COVID19-CT-dataset ¹³⁴	C, S	✓	✓							1000+
	NIH-NLST ¹³⁵	D	✓	✓	✓	✓	✓	✓	✓		~15,000
	LND ¹³⁶	C, S, D	✓	✓							294
	COVID-CT-MD ¹³⁷	C	✓	✓							308
Kidney CT	KITS 2019 ¹³⁸	S, D	✓	✓			✓	✓	✓		300
Bone CT	OAI ^c	D	✓	✓	✓						26,626,000
	HNSCC-3DCT-RT ¹³⁹	S	✓	✓							31
	Digital Hand Atlas ¹⁴⁰	L		✓	✓						1390
Body CT	DeepLesion ¹⁴¹	D	✓	✓							32,735
Fundus images	ODIR2019 ^d	C	✓	✓							5000
	AREDS2 ¹⁴²	C	✓	✓	✓	✓					4203
	PAPILA ¹⁴³	C	✓	✓							420
Eyes OCT	OCTAGON ¹⁴⁴	S	✓								233
	JHU-OCT ¹⁴⁵	S	✓								35
	OCT ¹⁴⁶	C	✓								384
Vessel ultrasound	TKTube ¹⁴⁷	S	✓	✓	✓					✓	233
Cardiac MRI	ACDC ¹⁴⁸	C, S, D	✓						✓		150
	Sunnybrook ¹⁴⁸	S, D	✓	✓							45
	UK Biobank ¹⁴⁹	S	✓	✓	✓						5903
Brain MRI/fMRI	ADHD-200 ¹⁵⁰	C	✓	✓						✓	200
	OASIS ¹⁵¹	C	✓	✓						✓	~500
	ABIDE ¹⁵²	C	✓	✓						✓	1112
	PPMI ¹⁵³	C, R	✓	✓						✓	254
	BraTS 2019 ¹⁵⁴	C, S	✓								335
	CANDI ¹⁵⁵	C, S, R	✓	✓						✓	103
	Cam-CAN ¹⁵⁶	C	✓								~650
	FCP ¹⁵⁷	C	✓	✓							/ ^e
	ADNI ¹⁵⁸	C, D	✓	✓						✓	800
	ABCD ¹⁵⁹	S	✓	✓	✓						4547
	UK Biobank ¹⁴⁹	S	✓	✓	✓				✓		22,528
Dermoscopy	ISIC ^{160,161}	C, S	✓	✓							10,000+
	Fitzpatrick-17k ^{162,163}	C			✓						16,577

^aC classification, S segmentation, D detection, L landmark detection, R registration.^bIncluding weight, height, and BMI.^c<https://nda.nih.gov/oai>.^d<https://odir2019.grand-challenge.org/>.^eThis dataset contains more than 1200 fMRI datasets collected independently at 33 sites.

Table 4 | A glossary of technical terms used in fairness research

Technical Terms	Definition
Adversarial learning	A type of neural network consisting of a target branch and an adversarial branch, aims to predict two variables, $\hat{Y}$ (class label) and $\hat{A}$ (sensitive attribute), respectively. However, although the loss function of the adversarial branch minimizes the prediction error of $\hat{A}$, the gradient is reversed by a gradient reversal layer ¹⁶⁴ , which pushes the network not to recognize A .
Disentanglement learning	The neural network extracts high-level features which is a mixture of information from the target task and the attribute. Disentanglement learning projects the feature vector into another space, where the information is split into independent components ¹⁶⁵ .
Contrastive learning	Contrastive learning takes paired samples from either the same class or different classes and uses loss functions that maximize the disparity between features from different classes and minimize the disparity between features from the same class ¹⁶⁶ .
Model calibration	Model calibration adjusts the predicted threshold of the model's output logits per subgroup, for example, regarding output logits larger than 0.65 (or 0.53) as the positive for the Male (or Female) subgroup, to ensure that the model has equal fairness criteria in each subgroup ⁶¹ .
Model pruning	For a pre-trained neural network, model pruning inspects all the neurons in the network, and removes some of them based on some fairness metrics, resulting in a network with fewer parameters ⁹⁰ .
Uncertainty	A variable that measures the stability of the neural network when facing the fluctuation of the input data ¹⁶⁷ .

Thus, it is important to discover the sources of unfairness in MedIA, not only to help clinicians understand potential biases in the AI algorithms and propose reliable treatment but also to help the AI scientists be aware of the source of unfairness and produce corresponding solutions and design better AI products. Moreover, after clinicians and AI scientists are aware of the sources of unfairness in each application, the government can devise actions focusing on the root reason and prevent the recurrence of unfairness in the future.

Following ref. 92, we categorize the sources of unfairness based on the components of a DL system pipeline, i.e., data, model, and deployment. Figure 7 illustrates the schema of these sources and potential solutions.

Unfairness existing in the data consists of three parts. *First*, the skewed data distribution of the diagnosis label and the sensitive attributes might cause the insufficient feature representation of the specific group^{4,93}. *Second*, due to the anatomical differences between different subgroups, the difficulty for DL models to provide precise diagnoses may also vary⁹⁴. *Third*, as the clinicians may have different annotation preferences on patients of different attributes⁹⁵, this annotation inconsistency will confuse the DL model and cause unfairness.

The DL model itself also causes unfairness. The general training process of DL models aims to find the model with the highest overall performance. However, a higher performance might lead to larger subgroup gaps⁵³. Besides, as DL models tend to learn easier but irrelevant correlations between the input image and the output diagnosis, they may attempt to produce the diagnosis based on spurious correlations rather than true medical evidence⁵³. Moreover, some DL models also amplify the unfairness that exists in the data and enlarge the fairness gaps among subgroups⁹⁶.

With the wide use of pre-trained DL models, unfairness also comes from the deployment. Some pre-trained models inherit bias from the pre-training datasets and perform unfairly on the downstream tasks. On the other hand, the non-neglectable domain gaps among the situations where the model is developed and deployed also affect the performance of the model and cause unfairness.

The Facial recognition (FR) community was aware of unfairness issues, even earlier than the MedIA community⁹⁷. Research in fair FR aims to achieve equal TPR and FPR of recognition among subgroups with different sexes, ages, hairstyles, emotions, etc. Over the past years, several attempts have been made to align fair MedIA with fair FR, by transferring unfairness mitigation algorithms from FR to MedIA^{53,54}. However, experiments prove that most of the useful methods in fair FR are not applicable in fair MedIA, which attracts people's attention to the differences between the two areas. Table 5 illustrates the comparison between fair FR and fair MedIA.

The largest difference between fair FR and fair MedIA is the variation of image modalities. As most of the images in FR are RGB images, the input in MedIA varies from 2D X-ray images, dermoscopy, and mammography to 3D MRI, and PET/CT. The huge disparity among the multi-modality input and complex types of tasks (classification, segmentation, detection) makes it impossible to find a common solution for all tasks. Besides, the amount of

samples in the MedIA dataset is several orders of magnitude smaller than those in facial datasets, which brings more difficulties for robust feature representation. The type of sensitive attribute also varies. While attributes in fair MedIA mainly describe the demographics of a patient, attributes in fair FR focus more on the appearance of a person. This disparity affects the difficulty of attribute recognition. As a result, the composition of MedIA datasets is usually skewed, due to the different morbidity across subgroups. In contrast, there are several specially designed face datasets with balanced attributes for fair FR, such as FairFace⁹⁷. The lack of balanced medical datasets seriously hinders the benchmark of unfairness mitigation algorithms in fair MedIA.

While AI scientists and clinicians have their understanding of fairness, the gaps between the mathematical definitions and the dilemma in medical scenarios cannot be neglected to achieve health equity.

Most of the current research in AI fairness is conducted on measuring and mitigating unfairness with the mathematical form of fairness, i.e. the numeric differences between manually designed metrics such as DP, AP, EqOdd, etc. However, from the scope of clinicians, *equality* in numbers does not always mean *equity* of treatment⁹⁸. In other words, the performance disparity does not equal unfairness. Forced insistence on numerical equality will destroy the causal relationship between the patients' metadata (sensitive attributes) and the diagnosis outcome.

To address fairness in AI, we need to decide which sensitive attributes should be evaluated. Generally, AI scientists prefer to analyze the statistical relationship between the target task and the sensitive attribute, i.e. does the model have a biased outcome on this attribute? On the contrary, clinicians pay more attention to the physiological causality between the two terms, for example, will the anatomical difference between the male and the female affect the diagnosis difficulty? This different paradigm of attribute selection also brings gaps between AI scientists and clinicians when they address fairness in MedIA together⁹².

The choices of fairness metrics also vary between AI fairness and clinical fairness. While AI scientists adopt metrics that are derived from the confusion metrics, clinicians regard some of the metrics as unreasonable. For example, one of the most commonly used metrics in AI fairness, demographic parity, requires that the patients from each subgroup should have the same probability of being predicted as ailing, i.e., $P(\hat{Y} = 1|A = 0) = P(\hat{Y} = 1|A = 1)$. However, in practical medical scenarios, where many illnesses are proven to be related to age or sex, the requirement of the same subgroup morbidity is not matched with reality. Furthermore, the diagnosis difficulty may vary across subgroups due to anatomical differences⁹⁹. In this situation, it is more appropriate for the DL algorithm to have an unequal diagnosis precision due to reasonable medical prior.

Moreover, due to the randomness of DL models, it is nearly impossible to have exactly the same performance among different subgroups. However, in medical applications, a slight fluctuation in the subgroup diagnosis performance is acceptable due to the complexity of the illness. Thus, we need to

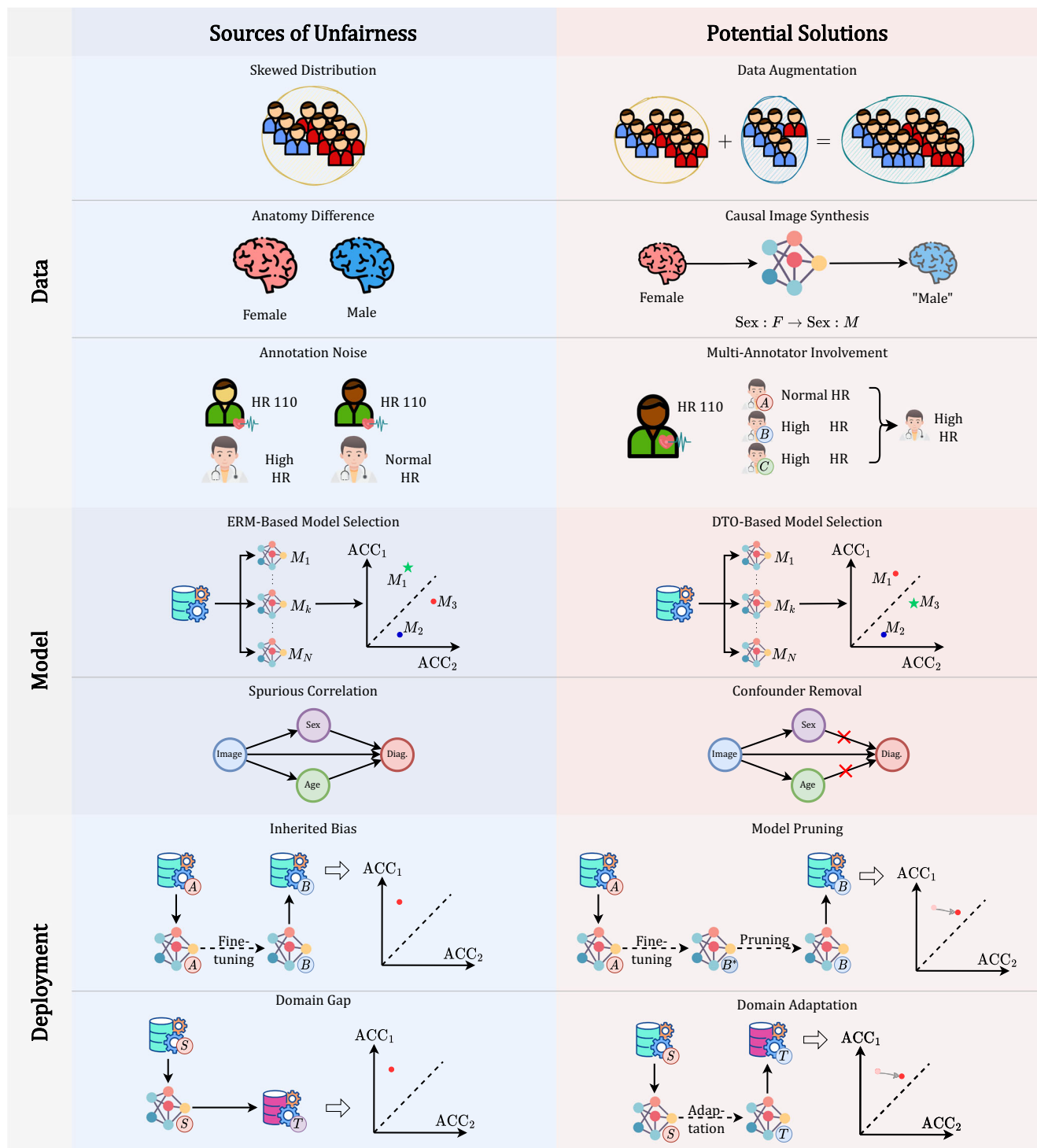


Fig. 7 | Sources of unfairness and potential solutions. From top to bottom: skewed data distribution → aggregate data from multiple datasets; anatomy difference between subgroups → ; annotation differences for each subgroup → using causal image synthesis methods to transfer the input to the synthesis ones with opposite attribute; annotation noise → involve multi-annotators to stabilize annotation; ERM-based model selection which chooses models with the highest overall

performance → DTO-based model selection which consider both performance and fairness; spurious correlations between sensitive attributes and diagnosis → removing the effects of the confounder; inherited bias from the pre-train dataset → pruning the pre-trained with fairness constraints; domain gaps between the source dataset and target dataset → using domain adaptation methods to transfer models.

carefully decide *what level of numerical difference means unfairness*. Furthermore, unlike utility metrics such as accuracy and area under the receiver-operating curve (AUROC), fairness metrics usually fluctuate along the training procedure and are hard to converge. Therefore, additional efforts need to be made to assess the level of fairness properly.

Recently, foundation models, such as Large Language Model (LLM)^{100,101}, Contrastive Language Image Pre-training (CLIP) and its variants^{102–105}, and Segment Anything Model (SAM) and its variants^{106–109} have attracted people's attention by their superior zero-shot or few-shot performances on downstream tasks in medical applications. However, as

Table 5 | Comparisons between Fair MedIA and Fair FR

	Fair MedIA	Fair FR
Image modality	2D: X-ray ⁵⁴ , dermoscopy ¹¹ , mammography ⁴⁵ 3D: MRI ³ , PET/CT ⁴⁶	RGB images ⁹⁷
Amount of samples	Range from tens to tens of thousands	Range from thousands to millions
Sensitive attributes	Sex, Age, Race, Skin tone, etc. (describing demographics)	Sex, Age, Race, Skin tone, Hair Color, etc. (describing appearance) ¹⁶⁸
Dataset composition	Skewed attribute distribution	Skewed attribute distribution: CelebA ¹⁶⁸ Balanced attribute distribution: FairFace ⁹⁷

the training sets of these foundation models are usually inaccessible, it is important to ensure whether these models have unbiased utilities on different subgroups before adopting them in healthcare applications.

For example, LLMs, a category of DL models trained with countless corpus from all over the world, might introduce unfairness from the pre-training tasks, or perform unfairly due to the large domain gap between the pre-training task and the fine-tuned downstream tasks (unfairness from deployment)¹¹⁰. Similarly, unfair performances are also witnessed in CLIP models, which try to align semantic information between text and image data to construct an excellent feature extractor^{111,112}. This unfairness might be due to the spurious relations between sensitive attributes and the target label¹¹³. SAM is another family of foundation models that focuses on image segmentation tasks, mainly with the help of point or box prompts. As shown in the original SAM paper, the training images of SAM have a strong region preference, i.e., the number of images collected in Europe is larger than that in South America. This bias in geographic distribution is propagated in the fine-tuned version of SAM in medical applications.

In short, most of the large-scale foundation models and language models suffer from different levels of unfairness, due to domain gap, annotation noise, spurious correlation, or inherited bias from the training set. This phenomenon hurts the trustworthiness of DL-based algorithms and must be handled properly. However, due to the huge amount of parameters of these foundation models, it is hard to mitigate unfairness using the aforementioned techniques, as the retraining of foundation models requires heavy computation and a long time. Thus, it is crucial to come up with unfairness mitigation methods aimed at foundation models, including perturbing the feature space¹¹⁴ and editing the input images¹¹⁵.

Addressing fairness in MedIA requires in-depth cooperation among AI scientists, ethicists, and clinicians. In this cooperation, AI scientists should be aware of the limitations of the mathematical form of fairness with the help of ethicists and clinicians, and try to develop new algorithms that can mitigate unfairness effectively. The clinicians could discover the causal relations between illness diagnosis and the metadata of the patients, finding out which type of disparity should be regarded as differences rather than unfairness. Besides, the government can also incorporate fairness considerations into clinical AI guidelines to improve the awareness of fairness in the whole pipeline¹¹⁶. By improving the comprehension of the operation mechanism of the DL models and the clinical context grasping, the researchers can update existing AI governance schemes to promote fairness. While establishing multi-directional communication across various professions may present challenges, it is imperative to exert considerable effort toward addressing fairness in MedIA and safeguarding health equity on a global scale.

In conclusion, this review reveals the importance of assessing fairness in deep learning-based medical image analysis. We describe the basics of group fairness and categorize current studies in fair MedIA into fairness evaluation and unfairness mitigation. Moreover, we address the challenges and opportunities for improving fairness in MedIA. In a word, fairness evaluation and unfairness mitigation is a rapidly growing and promising research field in MedIA, for AI scientists, clinicians, and the power-holder. We must pay more attention to this area to build a fair society for citizens of different sexes, ages, races, and skin tones.

Methods

This review was conducted based on the PRISMA guidelines¹¹⁷.

Search strategy

All the candidate articles were collected by a comprehensive search of four databases, including Scopus, PubMed, arXiv, and Google Scholar using the query conditions: ‘medical fairness’ | ‘fairness in medical image analysis’ | ‘fair medical machine learning’ | ‘fairness in healthcare’. We limit the publish time from 2015 to 2023 for Scopus and PubMed, and only include the top 200/100 items for ArXiv and Google Scholar, respectively.

After removing duplicated papers, we included all research papers in English. Since this review mainly focuses on methodology, we included papers describing methods for fairness issues in medical image analysis using deep learning algorithms.

For better categorization, we hierarchically split the papers into two classes, fairness evaluation and unfairness mitigation. Each of them was separated into more precise sub-areas accordingly. Besides, we also reported the image modality and datasets used in their paper, tasks and sensitive attributes they worked on, and metrics about fairness they adopted.

Data extraction

For the included studies, we extract information from the following aspects: (1) year of publication; (2) image modality; (3) datasets used; (4) type of task; (5) research area according to taxonomy; (6) sensitive attributes assessed on; (7) metrics used for fairness evaluation. By extracting these data, we hope to help the researcher better understand the routine of new studies about fair MedIA and come up with insights into both fairness evaluation and unfairness mitigation for MedIA.

Data availability

The authors declare that all data supporting the findings of this study are available within the paper.

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Author contributions

Z.X., J.L., and S.K.Z. designed the study. Z.X. developed the search strings and analyzed the data. Z.X. and J.L. wrote the first draft of the manuscript. H.L., Q.Y., and M.Z. extended and revised the “Discussion” section. All authors revised the manuscript and approved the final version of the submitted manuscript. The corresponding author attests that all listed authors meet authorship criteria and no others meeting the criteria have been omitted.

Competing interests

The authors declare no competing interests.

Additional information

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